SEC 4 · LEC 3 · ELECTRIC PROPULSION + ION THRUSTERS

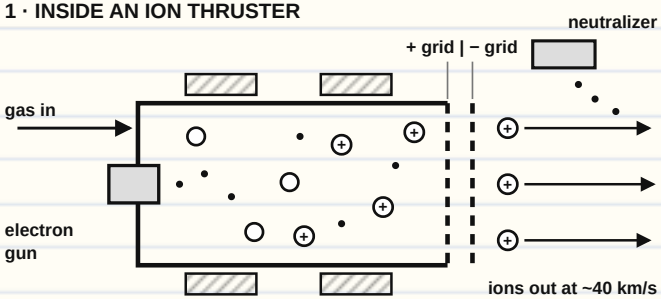
The tortoise engine

Left = your left notebook page. Right = your right page. Copy the drawing in pencil first, then the cues and notes. The summary box stays empty until the video and this PDF are closed.

LEFT PAGE · DRAW

S4 / L3 — Electric propulsion **date · pg __**

1 · INSIDE AN ION THRUSTER

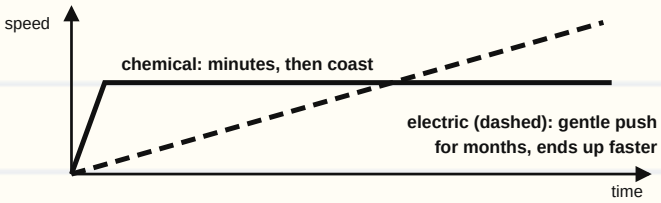


○ neutral atom ⊕ positive ion • electron
hatched = magnet · dashed = grid with holes

2 · THE FOUR STEPS

1 FEED xenon gas, a few mg per second	**2 IONIZE** electrons knock electrons off
3 ACCELERATE grids pull the ions out fast	**4 NEUTRALIZE** add electrons back to the beam

3 · TORTOISE AND HARE



✏️ **PENCIL KEY** · ○ atom · ⊕ ion · • electron · hatched = magnet · dashed line = grid · dashed curve = electric

⭐ **3 TAKEAWAYS** 3 lines

1. Electric = very fast exhaust, very little propellant, very small push.
2. Thrust is capped by electric power; it wins by pushing for months.
3. Ion thruster: ionize the gas, grids fling the ions out, neutralizer adds electrons.

🧠 **SUMMARY · video + PDF closed** 3 lines

1. _____
2. _____
3. _____

RIGHT PAGE · CORNELL

CUES (questions)	NOTES (keywords)
What do electric and chemical engines have in common?	electric propulsion : stored propellant → high-speed jet (same idea as chemical)
Why does electric use so little propellant?	Isp very high (3,000+ s) → exhaust very fast → very little propellant
Why is the thrust so low?	thrust low : limited electric power → tiny mass flow
How does a tiny push reach a high speed?	low thrust × very long time = very high final speed
What is it used for? What can't it do?	mature tech · most new satellites carry it uses: orbit changes (primary propulsion) · orbit maintenance · NOT launch or landing (no big thrust)
Ion thruster: the four steps?	ion thruster = accelerates ions with electricity
What do the two grids do?	ionize : electron gun fires electrons at neutral atoms → knocks an electron off → positive ion · plasma = ionized gas · magnets help
Why the neutralizer?	accelerate: voltage between + grid and − grid shoots ions out at ~40 km/s
Two example missions?	neutralizer : fires electrons into the beam → ship stays uncharged Deep Space 1 (NASA, 1998) · BepiColombo (ESA + JAXA, 2018 , to Mercury)
? what exactly is plasma	

SEC 4 · LEC 3 · TEACH

Static electricity, pointed backward

Simple story first, then the official names, then how it actually works with numbers you can check.

🧒 Like you're 5

Rub a balloon on your hair and it pulls your hair toward it without touching. That pull is electricity grabbing things that are **charged**. An ion engine is that trick, aimed out the back. Step one: take a gas and knock one tiny piece (an electron) off each atom. The atom is now charged, and it is called an **ion**. Step two: put two metal screens full of holes at the back, one charged plus and one charged minus. They yank the ions through the holes and fling them out at 40 km a second, forty times faster than a rifle bullet. Step three: squirt the knocked-off electrons out as well, or the ship would end up charged and pull its own exhaust straight back.

The push is about the weight of two sheets of paper. But it never stops. A chemical rocket is the hare: one huge sprint, then it is out of breath. The ion engine is the **tortoise**: a tiny push, every second, for years, and it ends up going faster on a fifth of the propellant.

🏷 The official names (name first, then what it means)

- **Electric propulsion, EP** (**solar electric propulsion, SEP**, when solar panels supply the power): electricity, not fire, accelerates the propellant.
- **Gridded ion thruster** (electrostatic ion thruster, ion engine): the type in this lecture. Its cousin the **Hall-effect thruster** is the most common electric engine today and was not covered.
- **Ionization** (here: electron-impact ionization): knocking an electron off a neutral atom. It costs the **ionization energy**, 12.1 eV for xenon.
- **Ion**: an atom with a net charge. Lose an electron and it is a positive ion.
- **Plasma**: gas hot or energetic enough that it is a mix of ions and free electrons. Overall it is still neutral. Often called the fourth state of matter.
- **Discharge chamber** (ionization chamber): the can where the gas is ionized.
- **Discharge cathode** (hollow cathode; the slide's "electron gun") and **anode**: the negative source of electrons and the positive wall that attracts them.
- **Potential difference** (voltage): the electrical "height" between two points. Charges fall through it and speed up.
- **Ion optics**: the pair of grids. **Screen grid** (the slide's "positive grid") and **accelerator grid** ("negative grid").
- **Neutralizer** (neutralizer cathode; the slide's "neutralizing electron gun"): emits electrons into the ion beam.
- **Electron backstreaming**: outside electrons rushing back into the engine. The negative grid blocks it.
- **Recombination**: an ion and an electron rejoining into a neutral atom.
- **Primary propulsion** / **station-keeping** / **attitude control**: the main Δv job / holding an orbit / pointing.

🛠 How it actually works

Step	The mechanism, with numbers (Deep Space 1's engine)
1 · Feed	Xenon gas trickles into the chamber: about **3 milligrams per second**. Xenon because it is heavy, easy to ionize and does not react with anything.
2 · Ionize	The cathode emits electrons; about 30 volts speeds them up. One that hits a xenon atom hard enough (12.1 eV) knocks an electron off: now it is a positive ion. Magnets make the electrons spiral, so they travel far farther before reaching the wall and get many more chances to hit an atom.
3 · Accelerate	Screen grid at **+1,100 V**, accelerator grid at −180 V, half a millimetre apart. An ion that drifts into a hole falls through that voltage: energy = charge × voltage = ½mv². For xenon, $v = \sqrt{(2qV \div m)} \approx$ **40 km/s**.
4 · Where the push lands	The grids pull the ions backward-out; the ions pull the grids forward with the same force. Newton's third law again. The thrust is felt on the grids.
5 · Neutralize	The beam carries off 1.76 amps of positive charge. Unbalanced, the ship would charge negative in microseconds and drag its own ions back: no thrust. The neutralizer emits the same 1.76 A of electrons.
6 · Thrust	$F = \dot{m} \times v_e \approx$ 3 mg/s × 30 km/s = **0.092 N**, 9 grams of push. (Effective exhaust is 30, not 40 km/s, because some gas leaves un-ionized.) Isp ≈ **3,100 s**.
7 · Power is the limit	That 0.092 N took 2,300 W, a whole solar array. Thrust ≈ 2 × efficiency × power ÷ v_e. For fixed power, faster exhaust means *less* thrust. About 40 mN per kW.
8 · Add it up	0.092 N on 486 kg adds 16 m/s every day. Deep Space 1 thrust for 678 days in total: Δv **4.3 km/s** from 74 kg of xenon. A chemical engine would need about 360 kg.

SEC 4 · LEC 3 · FACT CHECK

Where it is used, and what he got loose

The uses, the caption errors (many this time) and where the instructor was loose or wrong.

🛰️ What electric propulsion does and doesn't do

Job	Electric?	Why
Launch from the ground	Never	Thrust is 10,000+ times less than the craft's weight, and the engine only works in vacuum.
Landing, capture burns	No	They need km/s within minutes.
Raising orbit to GEO	Yes	Takes about 6 months instead of days, but saves tonnes of propellant.
Station-keeping	Yes, since the 1990s	Small, steady corrections: exactly what it is good at.
Deep-space cruise	Yes	Months of thrust add up to many km/s: Deep Space 1, Dawn, BepiColombo, Psyche.
Fine pointing	Starting to	Tiny electric thrusters already do this on a few science missions.

🔎 Caption errors

Captions said	Instructor meant
"stort much propellant… propeller"	**stored mass propellant**… propellant
"specific Kimballs" / "G-0 GZA"	specific **impulse** / **g_0**
"X and velocity", "section velocity"	**ejection** velocity
"lexing propulsion"	**electric** propulsion
"healthy object"	type of **objective**
"looking for parts and systems"	**launch or landing** systems
"I am now using a new tool Gus"	**ionizing a neutral gas**
"election", "electron gamma ray"	**electron**, electron **gun**
"a cathode and that out"	a cathode and **an anode**
"the Greeks", "negative grade"	the **grids**, negative **grid**
"the plastic"	the **plasma**
"in the camps the organization efficiency"	**enhance the ionization** efficiency
"audience", "electrons annoyance"	**ions**, electrons **and ions**
"a sound yak's a collaboration… All of them be"	**ESA and JAXA** collaboration… **BepiColombo**

▲ Where the instructor was loose

- **"Electrons collide with atoms breaking their electronic bonds."** No bonds are involved; xenon is single atoms. The collision knocks an electron *off the atom*. That is ionization.
- **"Plasma is… all of these particles which are positive."** Plasma is the ions *and* the free electrons together, neutral overall. Only the ions get pulled out by the grids.
- **"The neutralizer… so the gas will freely disperse."** The real reason is charge. Fire out only positive ions and the ship goes negative and pulls them back. This is the spacecraft charging from Section 2, self-inflicted.
- **"It cannot be used for launch… because of very low mass ejection."** Closer: its thrust is thousands of times smaller than the craft's weight, and it needs vacuum to run.
- **"By 2020 half of all satellites will have electric propulsion."** An old forecast that came true and more. Every Starlink has an electric thruster, and they are most of what gets launched.
- **BepiColombo "to perform orbital maneuvers around Mercury."** Its ion engines are for the trip *to* Mercury, braking against the Sun's pull for years. That module is dropped before arrival; chemical thrusters do the capture. Arrival was planned for November 2026, which is right now. I can't confirm whether it has happened.
- **"Electron gun" and "positive / negative grid"** are the diagram's labels. The engineering names are discharge cathode, screen grid and accelerator grid.
- **Not said: why low thrust.** He says the mass flow is low. The cause is power: every milligram needs about 500 joules to reach 30 km/s, and the panels only supply so many joules per second.
- **Not said: the Hall thruster**, the most common electric engine flying.

📒 Question bank cards

- Four steps of an ion thruster, in order.
- What limits an electric engine's thrust: propellant, or power?
- What happens if the neutralizer fails?
- Where on the engine does the push actually land?
- Name three jobs electric propulsion does and two it can't.

SEC 4 · LEC 3 · VOCABULARY

Every word used, in plain English

45 entries. This lecture is full of electricity words, so they are all here. "Repeat" = you have met it before.

Electric propulsion (EP) — engines that use electricity to accelerate propellant. *Repeat, L1.*

Solar electric propulsion (SEP) — electric propulsion powered by solar panels. Nearly all of it so far.

Stored mass propellant — the propellant carried on board. Here xenon gas. *Repeat.*

Propulsive jet — the stream leaving the engine. Here a beam of ions. *Repeat.*

Specific impulse (Isp) — exhaust speed ÷ g_0, in seconds. Electric: 1,500–4,000+ s. *Repeat.*

Ejection velocity = exhaust velocity (v_e) — how fast the propellant leaves. 30–40 km/s here. *Repeat.*

g_0 — 9.81 m/s², a constant. *Repeat.*

Thrust / low thrust — the push / here, a fraction of a newton. *Repeat.*

Electric power (watts, kW) — energy per second supplied by the solar panels. 1 kW = 1,000 W.

Mature technology — proven, flown many times, no longer experimental.

Orbit change / primary propulsion — the engine that provides the mission's main Δv.

Orbital maintenance = station-keeping — small burns that hold an orbit against drift. *Repeat, scene 1.*

Attitude control — controlling which way the spacecraft points. *Repeat. Section 6.*

Launch / landing systems — getting off a planet / down onto one. Both need thrust bigger than weight.

Peak thrust — the most push an engine can give at one moment.

Ion thruster (gridded ion engine) — ionizes gas and accelerates the ions with charged grids. *Static electricity, pointed backward.*

Atom / neutral atom — the smallest piece of an element / one with equal + and − charges, so no net charge.

Electron — the tiny negative particle around an atom. Easy to knock off.

Ion / positive ion — an atom with a net charge / one that lost an electron.

Ionize / ionization — to turn atoms into ions / the process.

Ionization energy — the energy it takes to knock one electron off. Xenon: 12.1 eV. *eV is a repeat from Section 2.*

Charge (positive / negative) — the property electricity acts on. Opposites attract, likes repel.

Electric field — the region where a charge feels a push. Between the grids it is very strong.

Potential difference = voltage (V) — electrical 'height' between two points. A charge speeds up falling through it. *A hill for charges.*

Cathode / anode — the negative electrode that gives off electrons / the positive one that collects them.

Electron gun — the slide's name for the cathode that sprays electrons into the chamber.

Electron beam — a stream of electrons.

Collide / collision — hit. Here an electron hitting an atom.

Plasma — ionized gas: ions and free electrons mixed, neutral overall. Lightning, neon signs, the Sun. *The fourth state of matter.*

Discharge chamber — the can where the gas is ionized.

Magnet / magnetic field — makes electrons spiral so they hit more atoms.

Ionization efficiency — what fraction of the gas actually gets ionized.

Grid — a thin metal plate with thousands of holes.

Screen grid = positive grid — the inner grid, at high positive voltage.

Accelerator grid = negative grid — the outer grid, slightly negative. Pulls ions out and keeps outside electrons from getting in.

Repel — push away. Negative repels negative.

Neutralizer = neutralizing electron gun — a small cathode that sprays electrons into the ion beam so the ship stays uncharged.

Neutralize / charge balance — make the + and − equal again.

Recombination — an ion and an electron joining back into a neutral atom, out in the beam.

Disperse — spread out and drift away.

Xenon — the heavy, inert gas ion engines use. *Repeat, L1.*

Ampere (A) — the unit of electric current: charge flowing per second.

Hall-effect thruster — the other main electric engine; not in this lecture. *Repeat, L1.*

Deep Space 1 — NASA's 1998 test probe, first to cruise on an ion engine. *Who's Who.*

BepiColombo / ESA / JAXA — Mercury mission launched 2018 / European and Japanese space agencies. *Who's Who.*

SEC 4 · LEC 3 · COPY INTO NOTES

Where each piece goes in the notebook

Routes A to D for this lecture. The concept cards (route E) are on the next page.

A · Equations & key numbers page (p3–4)

$v = \sqrt{(2qV \div m)}$	ion speed from grid voltage
	xenon, 1,100 V → 40 km/s
$F \approx 2 \cdot \eta \cdot P \div v_e$	thrust from power (η = efficiency)
	2,300 W → 0.092 N · ~40 mN per kW
ion engine	Isp ~3,100 s · $\dot{m}$ ~3 mg/s · 16 m/s per day
Deep Space 1	678 days thrust · 74 kg Xe · Δv 4.3 km/s

B · Section 4 map page (add to the sentence)

```
electric: ionize xenon → grids (+1,100 V) fling ions at 40 km/s
→ neutralizer adds electrons; tiny push for years, power-limited
```

D · Question bank ("but how?" cards)

```
Q: How does an ion engine speed the gas up with no fire?
A: Knock an electron off each atom (ion). Charged grids pull the
   ions through 1,100 V: v = √(2qV/m) ≈ 40 km/s.
Q: Why does it need a neutralizer?
A: Positive ions leave, so the ship would charge negative and pull
   them back. Equal electrons out keeps it neutral.
Q: Why is the thrust so small?
A: Power. Fast exhaust costs ~500 J per milligram; panels give
   only a few thousand joules per second.
Q: How can 9 grams of push get anywhere?
A: It never stops: 16 m/s per day, 4.3 km/s after 678 days.
```

C · One-line word bank (new lines)

```
p__ · electric propulsion (EP) = SEP · electricity throws it  → C20
p__ · primary propulsion · the engine doing the main Δv
p__ · ion thruster = gridded ion engine · ionize, then grids  → C20
p__ · atom / electron · smallest bit of an element / its − particle
p__ · ion · atom with a net charge (lost an electron = +)
p__ · ionization · knocking an electron off an atom
p__ · ionization energy · cost of that: xenon 12.1 eV
p__ · plasma = ionized gas · ions + electrons, neutral overall  → C21
p__ · charge · what electricity acts on; opposites attract
p__ · voltage = potential difference · a hill charges fall down
p__ · cathode / anode · gives off electrons / collects them
p__ · electron gun = discharge cathode · sprays electrons in
p__ · discharge chamber · the can where gas is ionized
p__ · screen grid (+) / accelerator grid (−) · pull the ions out
p__ · neutralizer · electrons into beam; ship stays neutral → C21
p__ · electron backstreaming · outside electrons rushing in
p__ · recombination · ion + electron → neutral atom again
p__ · Hall-effect thruster · other electric engine, magnetic
p__ · ampere (A) · current: charge per second
```

SEC 4 · LEC 3 · CONCEPT CARDS

Route E: pages C20 and C21

Two notebook pages, two cards each. Copy line for line; the faint numbers are your ruled lines.

WORD BANK · CONCEPT CARDS · scene 9 — C20

ELECTRIC PROPULSION (EP)

= electricity, not fire, accelerates the propellant

WHY chemistry caps exhaust at ~4.5 km/s. Electricity has no cap, so EP uses a tenth of the propellant

\# Isp 3,000+ s · thrust under 1 N · runs for years

≠ a weak = bad engine (more total Δv)

→ power limits thrust · ion + Hall thrusters

ION THRUSTER = gridded ion engine

= ionizes gas, then grids fling the ions out

WHY a charged atom can be grabbed by voltage and thrown far faster than any flame can throw gas

\# +1,100 V → xenon ions at 40 km/s · 0.092 N from 2.3 kW

≠ Hall thruster (magnetic, no grids)

→ ionize → accelerate → neutralize

WORD BANK · CONCEPT CARDS · scene 9 — C21

PLASMA = ionized gas

= ions + free electrons, neutral overall

WHY neutral gas ignores electricity. Knock electrons off and you can steer and accelerate it

\# xenon: 12.1 eV per atom to ionize

≠ 'only positive particles' (electrons are in it too)

→ ionization · ionosphere · spacecraft charging

NEUTRALIZER = neutralizer cathode

= sprays electrons into the ion beam

WHY shoot out only + ions and the ship goes negative and pulls them back: thrust dies

\# Deep Space 1: 1.76 A of ions out, 1.76 A of electrons out

≠ the electron gun inside (that one ionizes)

→ spacecraft charging · recombination

C20: 11 + 10 lines of 22 · C21: 11 + 10 lines of 22

SEC 4 · LEC 3 · SAY IT OUT LOUD

Last page: prove it by talking

After the spread and the 3-line summary, with everything closed. Send me what you said and I will grade it like a friend would.

🗣 The friend's opening question

"An engine that pushes as hard as two sheets of paper? How is that any use?"

✅ Three sentences you should be able to say

- Because it never stops. It adds a few metres per second every day for years, and ends up faster than a chemical rocket on a fifth of the propellant.
- It works like static electricity: knock an electron off each gas atom so it is charged, then two charged grids fling those ions out the back at 40 kilometres a second.
- The catch is power. Throwing gas that fast takes a lot of electricity and solar panels only give so much, so the push is tiny and you can't launch or land with it.

❓ Follow-ups a friend might ask

- "What's an ion?" (an atom that lost an electron, so electricity can grab it)
- "What's plasma?" (gas where the atoms have been split into ions and loose electrons)
- "Why spit the electrons out too?" (otherwise the ship charges up and pulls its exhaust back)
- "Has it actually flown?" (since 1964; Deep Space 1 in 1998, Dawn, BepiColombo, every Starlink)
- "Why not just add more solar panels?" (you can, but panels are mass, and far from the Sun they fade)

📝 Your explanation (paste or dictate)

_ send it to me in chat; I'll put it here _

🎯 My read

solid on: _
fuzzy on: _
fix: _
→ question-bank cards: _

⚡ Two lectures left in Section 4. Send them the same way.